

OFFICIAL ACSEF ABSTRACT FORM

The Abstract is a required part of your project. Bring 11 copies to Project Set-up before the Fair.

One copy will be displayed. You may bring a free standing clear photo frame or bring a page protector & blue painter's tape so ACSEF officials will tape it to the front of the display table. Other copies will be for judges & public visitors to take away.

Your abstract must be written **after** data is collected, analyzed, and conclusions are made.

- Type directly into the PDF form. Save to your computer when finished..
- Type your project ID in the upper left corner of the large blank space below (ID begins with MS- or HS) •
Type your project title directly beneath the project ID
- Type your AI nonuser or user statement below the project ID
- Press return ONE time - creating a blank line underneath the AI statement.
- Copy and paste your word document abstract into the remaining space
- Answer the Certification Statements at the bottom.

Type using **11 point, black font**, single spaced. Your abstract should be **250 words or less and fit within the frame below.**

HS-GEOC-131

Exploring Innovative Machine Learning Models for Soil CO₂ Emission Forecasting

AI was not used in creating this display item

Soil respiration, the natural process by which carbon dioxide (CO₂) is released from the soil into the atmosphere, is the largest flux of CO₂ from the Earth's surface. Despite its significance, accurately predicting CO₂ emissions caused by soil respiration remains challenging, and the influence of environmental variables are not yet fully understood. Early studies using artificial intelligence (AI) to predict soil respirations often relied on linear models or overly simplified datasets which lacked real-world applications. This study introduces a novel integration of XGBoost and Shapley Additive exPlanations (SHAP) to achieve both high predictive accuracy and detailed feature interpretability. In this study, an innovative ensemble model, XGBoost, is developed to estimate annual soil respiration (Rs) using spatial, temporal, climatic, and other independent variables. Additionally, a deep neural network model is explored. Among the machine learning models evaluated, XGBoost shows the best performance, with a 4% reduction in Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) and a 3.7% increase in R² compared with the Random Forest model, which was so far believed to be the state-of-the-art method. To aid in model explainability, SHAP is used to give feature importance at both the global and individual levels of analysis. Results indicate that Mean Annual Temperature (MAT) and Mean Annual Precipitation (MAP) are the most dominant environmental factors, with geographic variables such as latitude and longitude also significant. This study fills the gap between predictive modeling and interpretability, providing an explainable AI framework that is scalable for real-world environmental research.

Certification Questions: Answer each either YES or NO

1. As a part of this research project, the student directly handled, manipulated, or interacted with human participants, vertebrate animals, potentially hazardous biological agents:(including PHBA's such as: microorganisms, rDNA, tissues.

N

2. I/we worked or used equipment in a regulated research institution (college or professional lab) or industrial setting.

N

3. This project is a continuation of previous research. **N**

4. My/our display board includes non-published photographs/visual depictions of humans (other than myself). **N**

5. This abstract describes only procedures performed by me/us, reflects my/our own independent research, and represents one year's work only. **Y**

6. I/we hereby certify that the abstract and responses to the above statements are correct and properly reflect my/our own work. **Y**